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Coral Reefs
Journal of the International Society for Reef Studies

ISSN 0722-4028

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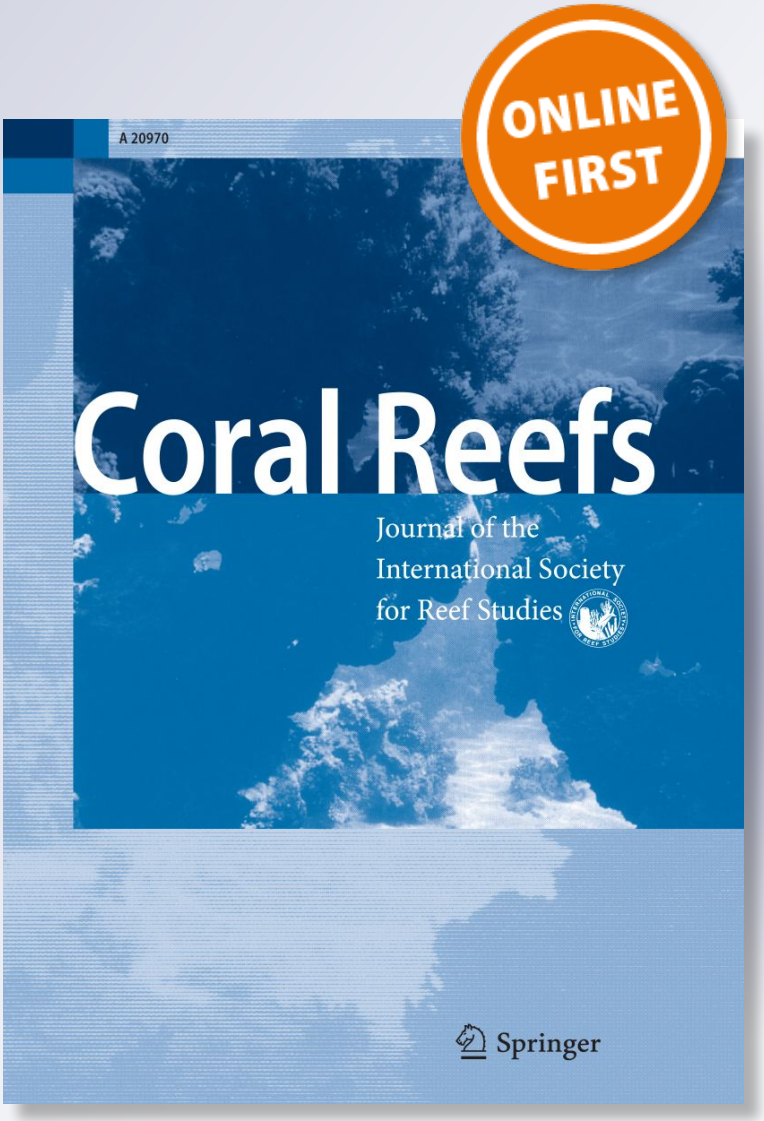

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https://doi.org/10.1007/s00338-019-01789-6

REPORT

Sustained mass coral bleaching (2016–2017) in Brazilian turbid-zone reefs: taxonomic, cross-shelf and habitat-related trends

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Received: 11 August 2018 / Accepted: 7 March 2019
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Abstract Between 2014 and 2017, an unprecedented heat stress accumulated and propagated across the tropical oceans and resulted in the so-called Third Global Bleaching Event (TGBE). Information about the effects of the TGBE in marginal coral reef provinces are still scarce, but can be relevant to understand the trajectories of coral reefs as climate changes intensify. Akin to deep mesophotic reefs and reefs in thermally stressed regions, low diversity, turbid-zone reefs may exhibit high bleaching tolerance due to local adaptations and conditions (e.g., shading by turbidity). Here, we summarize previous bleaching events in the tropical Western South Atlantic Ocean and explore taxonomic, cross-shelf and habitat-related bleaching trends in the Abrolhos reefs in February, May, June and October 2016, and March 2017. Fire corals (*Millepora* spp.) were the most affected, but all scleractinian species and several octocorals and zoanthids also bleached. Bleaching prevalence was higher in shallow coastal and offshore reef arcs than in deeper mesophotic reefs. All coral species bleached, but there were taxonomic and habitat-related trends in bleaching prevalence. Several species bleached less in the sites and habitats where their abundance was lower. As of March 2017, coral mortality was overall low across the region (< 3% of total coral cover). Our results add to the recent evidence that deep reefs provide partial refugia for a few coral species, and that turbid-zone reefs may be less susceptible to climate stress due to shading, higher heterotrophy levels, and local adaptations.

Keywords Southwestern South Atlantic · Monitoring · Mesophotic reefs · ENSO · Disturbance

Topic editor Morgan S. Pratchett.

Electronic supplementary material The online version of this article (https://doi.org/10.1007/s00338-019-01789-6) contains supplementary material, which is available to authorized users.

Carolina D. Teixeira and Rodrigo L. L. Leitão have equal contribution.

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Introduction

Mass coral bleaching episodes have been reported in scattered localities since the early twentieth century (Verrill 1902; Vaughan 1914), and after the 1980s, these events reached regional to global scales (e.g., 1982–1983, 1987–1988, 1994–1995, 1997–1998, 2002–2003, 2015–2016) (Hughes et al. 2017). Global bleaching episodes often occur during positive phases of the El Niño–Southern Oscillation (ENSO) (Claar et al. 2018) and can be forecast based on the duration and intensity of sea surface temperature (SST) anomalies (Eakin et al. 2016). For the Western

Published online: 15 March 2019

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South Atlantic (WSA), bleaching episodes have been reported since 1993 (Migotto 1997), but systematic monitoring at the regional scale is lacking (see Ferreira and Maida 2006). The WSA encompasses turbid-zone reefs dominated by stress-tolerant species (e.g., Moura et al. 2016; Bastos et al. 2018). Akin to "marginal" reefs in the Indo-Pacific (van Woesik et al. 2012; Morgan et al. 2017) and the Caribbean (Wagner et al. 2010; García-Sais et al. 2017), WSA reefs are dominated by stress-tolerant species and may be more resistant and resilient to thermal anomalies. Such putative "nearshore climate-change refugia" are subjected to reduced irradiance due to dissolved and suspended matter, and their corals may be more adapted to higher temperatures typical of shallower coastal waters (van Woesik et al. 2012).

Bleaching episodes may have variable outcomes, from full recovery to mass mortality, depending on the intensity and duration of the environmental stress (Glynn 1996; Baker et al. 2008; Hughes et al. 2017), as well as on the overall health state of the ecosystem (Hoegh-Guldberg 1999). It is recognized that coral colony morphology is often related to bleaching susceptibility (Brown and Suharsono 1990; Gleason 1993), but the full array of drivers that lead to increased resistance and resilience is not fully understood. Resistance to bleaching may be associated with local conditions such as upwelling and shading (Chollett et al. 2010; Morgan et al. 2017), adaptation to extreme conditions (van Woesik et al. 2012), and depth – the "deep reef refugia hypothesis" (Bongaerts et al. 2010; Smith et al. 2014). In "marginal" reefs, associations with thermo-tolerant symbionts (Marshall and Baird 2000; Loya et al. 2001) and heterotrophic feeding (Anthony 2006) may also drive increased resistance and resilience to bleaching.

Understanding the interaction among causal factors and the consequences of mass bleaching episodes is critical to evaluating the health and forecast the trajectories of reef ecosystems. Here, we present a compilation of previous bleaching events in the WSA and describe the effects of the Third Global Bleaching Event (TGBE) (Eakin et al. 2016) in the region's largest coralline system, the Abrolhos reefs. We contrasted the susceptibility and short-term resilience of the most common corals and zoanthids, and explored habitat-level and cross-shelf bleaching trends across 2016 (February, May, June and October) and 2017 (March).

Materials and methods

Study area

The Abrolhos Bank is a 46,000 km^2 enlargement of the Eastern Brazilian shelf (16°40′–19°40′S, 39°10′–37°20′W) off Bahia and Espírito Santo states. The region encompasses the largest (> 8000 km^2) and richest reef complex in the South Atlantic (Laborel 1969; Moura et al. 2013). The Abrolhos reefs are unique for occurring as mushroom-shaped pinnacles ('chapeirões') built by bryozoans, coralline algae and corals under a low storm disturbance regime (Bastos et al. 2018). The unique shape of these pinnacles includes two contrasting habitats that occur in close proximity but encompass highly distinct benthic assemblages (Francini-Filho et al. 2008): tops (flat and well lit, 1–10 m depths) and walls (steep and poorly illuminated, 5–25 m depths) (Fig. 1). These emergent and quasi-emergent structures occur in two arcs (seabed at 20–25 m depth), one ~ 10 km offshore (coastal arc) and another ~ 55 km offshore (offshore arc) (Fig. 1). Coastal reefs are subjected to higher fishing and turbidity levels from terrigenous sources, while reefs in the offshore arc are less exposed to such land-based stressors (Bruce et al. 2012) and are within a no-take zone established in 1986. Deeper mesophotic reefs occur to the east of the offshore arc, either as columnar pinnacles that lack the mushroom shape (tops between 10 and 20 m depths) or as eroded carbonate platforms with 1–5 m heights (seabed at 30–90 m depths) (Moura et al. 2013; Bastos et al. 2018).

Sampling

Sampling sites were distributed in the coastal and offshore arcs (two sites in each arc) and also included one deeper mesophotic site (California Reef) between 12 and 30 m depth (Table 1). The high structural heterogeneity of the Abrolhos reefs has a substantial effect on benthic cover estimates obtained from different sampling approaches (Francini-Filho et al. 2013). Therefore, we used transects and photo-quadrats to estimate bleaching prevalence and to develop spatial and temporal contrasts. For spatial contrasts (habitat vs. arc) we sampled 2–3 line transects (100–150 m) at each habitat and locality (n = 18) during the peak of the bleaching event (May 2016). All colonies intercepted by the line were assigned to one of the following categories (Guest et al. 2012): (1) healthy (H), when colonies displayed normal coloration; (2) partially bleached (PB), when coloration was up to 50% lighter than H; (3) heavily bleached (HB), when bleaching was well evident and ranged from 50% lighter than H to completely white; (4) bleaching and mortality (BM), when tissue loss and/or biofilm/secondary colonization were recorded in all or part of the colony. Temporal variation was assessed from 10 randomly placed fixed photo-quadrats repeatedly sampled at each habitat (top and wall) and site (n = 5) in February, May, June and October 2016, and March 2017. Offshore arc sites (n = 2) were sampled with photo-quadrats in all periods, although coastal arc sites were not sampled in October 2016, and the deeper mesophotic site (California Reef) was sampled in May and June 2016, and March 2017.

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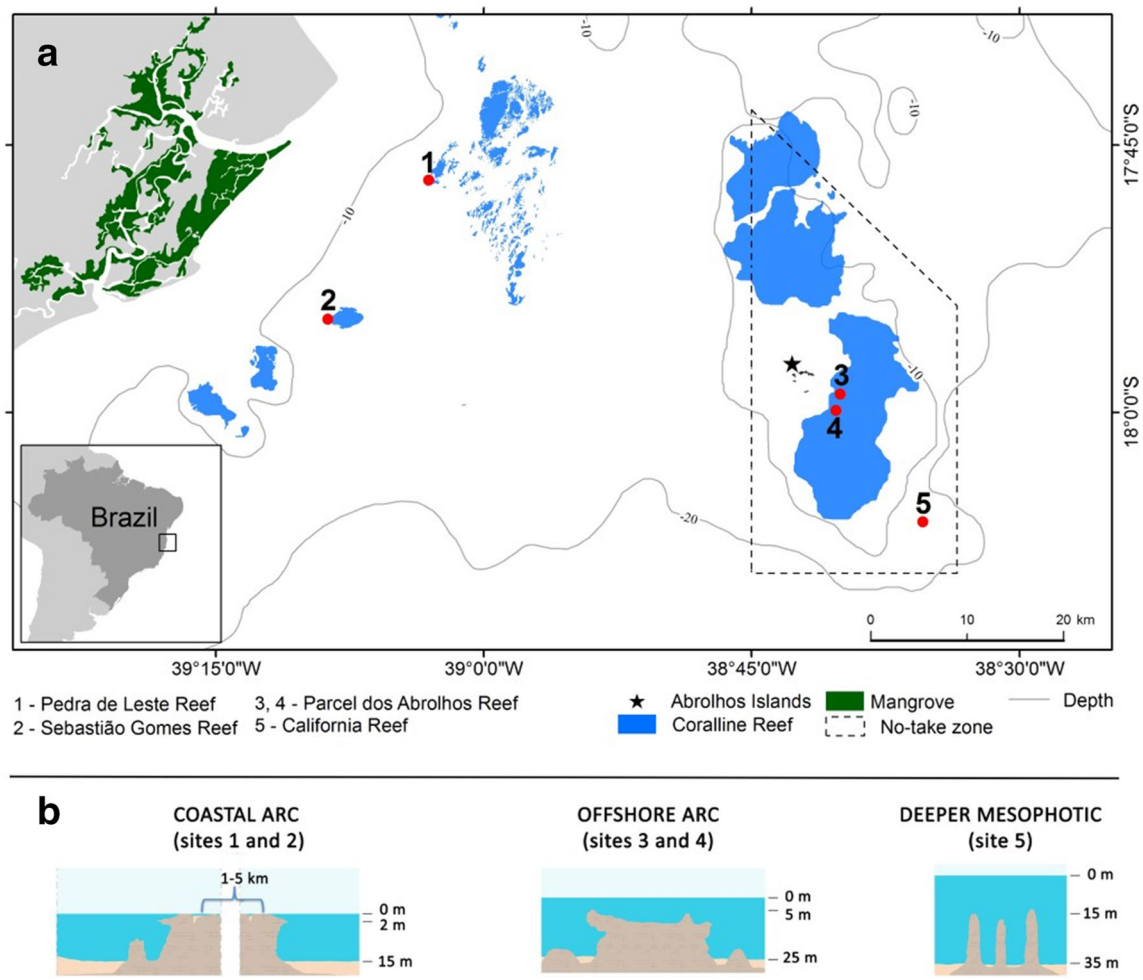

Fig. 1 The Abrolhos Reefs, Brazil. Map showing sampling localities (**a**) and the general morphology of the pinnacles at each cross-shelf stratum (**b**). Isobaths in meters

Table 1 Characteristics and coordinates of the sampling sites in the Abrolhos reefs, Brazil

Sampling site	Habitat/max depth (m)		Distance offshore (km)	Location (lat/long)
	Top	Wall		
Coastal arc				
Pedra de Leste reef (Site 1)	2.3	6.1	14.3	− 17.783/− 39.051
Sebastião Gomes reef (Site 2)	3.2	8.2	13.7	− 17.913/− 39.145
Offshore arc				
Parcel dos Abrolhos Reef (Site 3)	7.3	16.9	61.4	− 17.983/− 38.667
Parcel dos Abrolhos Reef (Site 4)	6.9	17.0	61.7	− 17.998/− 38.671
California Reef deeper mesophotic (Site 5)	25	30	70	− 18.102/− 38.590

Environmental variables

Monthly Multivariate ENSO Index (MEI) values (Wolter and Timlin 1993) were obtained in December 21 (2017), from NOAA's Earth System Research Laboratory (esrl.noaa.gov/psd/enso/mei). In situ daily mean temperatures were calculated from hourly records using HOBO Pendant® data loggers model UA-002-08 installed at pinnacles' tops (5–8 m depths) in coastal and offshore arc reefs (2016 and 2017), and at the deeper mesophotic site at

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20 m depth (July 2016–March 2017). Nighttime SST (4 μ, 5 km resolution) and the Degree Heating Week (DHW) index version 3 (Liu et al. 2014) were obtained in October 31st, 2017 from NOAA's Coral Reef Watch (CRW) (coralreefwatch.noaa.gov), using the Abrolhos' virtual station (18°S, 38.5°W). The CRW system uses satellite-derived nighttime SSTs to identify values of 1 °C above maximum monthly means ("Hot Spot") and quantifies accumulated thermal stress over 12 weeks, delivering an associated bleaching alert scheme (Liu et al. 2006). Diffuse attenuation coefficient for downwelling irradiance at 490 nm (KD490) from the MODIS/Terra sensor was obtained at monthly frequency and 9 km spatial resolution from NASA's OceanColor Web (oceancolor.gsfc.nasa.gov) and used as a proxy for turbidity. All remote sensing data were processed using SeaDAS software ver. 7.4 (seadas.gsfc.nasa.gov).

Data analyses

Analyses included the 10 most common taxa, out of the ~ 19 scleractinian and 4 fire corals (*Millepora* spp.) recorded in Abrolhos (Castro et al. 2005): *Agaricia* spp., *Favia gravida*, *Madracis decactis*, *Montastraea cavernosa*, *Mussismilia braziliensis*, *M. harttii*, *M. hispida*, *Porites* spp., *Siderastrea* spp. and *Millepora* spp. Relative cover (%) of each species was estimated from photo-quadrats and is presented as percentages of total cover of hard and fire corals. Each photo-quadrat consisted in 15 high-resolution contiguous images totaling 0.5 m^2 processed with the software Coral Point Count with Excel Extensions (Kohler and Gill 2006). Organisms' relative cover and bleaching categories were recorded using 600 randomly placed points per quadrat. The prevalence of the three first bleaching categories (H, PB and HB) for each species was estimated from transect data, while temporal trends were assessed with photo-quadrat data converted into a bleaching index (BI) following McClanahan (2004):

$$\mathrm{BI} = \frac{0 * \mathrm{H} + 1 * \mathrm{PB} + 2 * \mathrm{HB}}{2}$$

where H, PB and HB correspond to the bleaching level categories.

Spatial contrasts were explored with permutational multivariate analysis of variance (PERMANOVA; Anderson 2001) based on a Euclidian Distance matrix, using arc and habitat as fixed factors, and the BI as the response variable (no transformations applied). Analyses were carried out with PRIMER 6 + PERMANOVA software (Anderson et al. 2008).

Results

Five bleaching episodes were reported in the WSA since 1990, all associated with the previous global events (1998, 2010) and positive phases of ENSO (Fig. 2). Since 2000, the Abrolhos reefs were monitored during all austral summers, with lower than moderate bleaching (> 30–50% of colonies bleached) recorded up to 2016. Severe bleaching (> 50%) was recorded in 1993 and 1998 (see reference list in Fig. 2), with 9 and 5 affected species, respectively.

From 2014 to 2017, SSTs ranged from 24.2 to 29.6 °C, with highest DHW values between April and June 2016 (Fig. 3a). Satellite-derived temperatures above the regional bleaching threshold (27.9 °C) were frequent and resulted in 8.4 DHW (7 days > 8 DHW) and 6.7 DHW (11 days > 6.5 DHW) in May 2014 and 2015, respectively. In April 2016, the Abrolhos reefs reached 9.7 DHW (57 days > 8 DHW), and maximum values (47 days > 9 DHW) were sustained for 37 days. Despite the subsequent colder phase, from July 2016 on (maximum DHW = 4.0, April 2017), bleaching levels remained high at least until March 2017. In situ temperatures recorded in the coastal and offshore arcs varied similarly and showed the same trend and pattern from SSTs obtained from remote sensing (Fig. 3b). From July to September 2016, in situ temperatures from the deeper mesophotic reef also followed similar SST trends to the shallow reefs. However, from October 2016 to February 2017, in situ temperatures in the deeper mesophotic reef remained consistently lower than SSTs (Fig. 3b). Turbidity was significantly higher and more variable in the coastal arc (Fig. 3c).

Benthic cover in the coastal and offshore arcs was dominated by turf, macroalgae and CCA (crustose coralline algae, see Electronic Supplementary Material). Coral cover ranged from 9.4 to 39.8% in the coastal arc (values from pinnacles' tops and walls, respectively), and from 5.7 to 18.4% in the offshore arc (walls and tops, respectively; see Electronic Supplementary Material). The high coral cover in the coastal arc walls is related to the dominance (> 90%) of *Montastraea cavernosa*. The 2016–2017 bleaching episode was remarkable for its severity and for affecting all coral species (Table 2, Fig. 4), and also for being sustained across two summers, with bleached colonies recorded in February, May, June and October 2016, and in March 2017 (Fig. 4). Bleaching prevalence varied among species, sites, and along the study period. The most affected species on pinnacle's tops were *F. gravida*, *Millepora* spp., *M. cavernosa*, *M. harttii*, *M. hispida*, *Porites* spp. and *Siderastrea* spp., with > 40% of colonies bleached in at least one site during May 2016 (Tables 2 and 3, Fig. 5). On pinnacles' walls the most affected species (> 40%) were *Agaricia*

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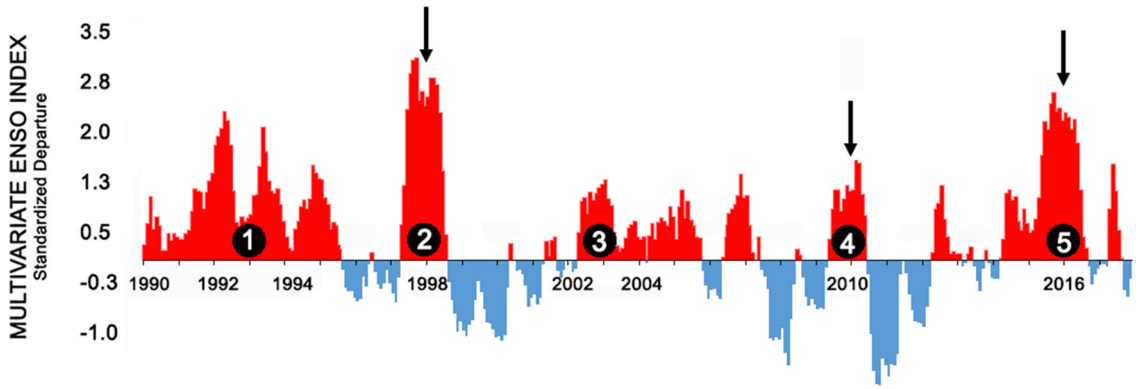

	1	2	3	4	5
Oceanic Islands	N.D.	N.D	X	X	N.D.
Eastern Brazil	N.D.	N.D.	X	5	N.D.
Northern Bahia	N.D.	7	2	9	N.D.
Abrolhos	9	5	0	0	10
Southeastern Brazil	2	N.D.	N.D.	N.D.	N.D.

Fig. 2 Multivariate ENSO Index (MEI) from 1990 to 2017 and the occurrence of coral bleaching. Numbers (1–5) within black circles indicate the number of bleaching events reported in Brazil during each MEI positive anomaly and arrows indicate the three global bleaching events. Numbers in the table indicate how many species were affected in each bleaching event and their respective location within the Western South Atlantic. *ND*, no data; *X*, bleaching reported, but the number of affected species is unknown Data compiled from Migotto (1997), Castro and Pires (1999), Dutra et al. (2000), Leão et al. (2003), Ferreira and Maida (2006), Kelmo and Attrill (2013), Ferreira et al. (2013), Miranda et al. (2013), Dias and Gondim (2016)

spp., *M. cavernosa*, *M. hispida*, *Porites* spp. and *Siderastrea* spp., the first two being more abundant in this habitat than in pinnacles' tops (Tables 2 and 3, Fig. 5). Conversely, *M. braziliensis* was less affected on tops (< 40%), and *M. harttii* and *M. decactis* were less affected on walls, where the latter is more abundant (Tables 2 and 3, Fig. 5). Six species (*Agaricia* spp., *F. gravida*, *Millepora* spp., *M. braziliensis*, *M. harttii* and *M. hispida*) presented significant between-habitat differences in bleaching prevalence (Table 3). Interactions between arc and habitat effects were detected only for *Agaricia* spp. and *M. harttii* (Table 3).

In the coastal and offshore arcs, the relationship between the Bleaching Index (BI) and the relative abundance of each coral species was variable (Fig. 5). For instance, the BI for *Agaricia* spp. and *M. hispida* was lower in sites and habitats where they were more abundant, while *F. gravida*, *M. decactis*, *M. cavernosa* and *M. harttii* presented an opposite trend (Fig. 5). Although we did not quantify bleaching in corals that represented < 1% of total coral cover, we recorded affected colonies of *Mussismilia lepthophylla* (PB), *Meandrina braziliensis* (HB) and *Scolymia wellsi* (PB and HB). In addition, the zoanthids *Palythoa caribaeorum* and *Zoanthus* spp. (PB, HB and BM), as well as the octocorals *Plexaurella grandiflora* and *P. regia* (HB, BM) were also affected.

In the deeper mesophotic site (California Reef), corals covered 14.8% of the substrate (13% scleractinian, 1.8% milleporans). The coral assemblage included nine species and was dominated by *M. cavernosa* (8.5% of total benthic cover). Six species bleached, but prevalence was lower than in the emergent reefs, not reaching the moderate bleaching status (> 30% prevalence). The most affected species in June 2016 were *Agaricia* spp. (3% PB), *M. cavernosa* (4.5% PB), *M. hispida* (6% PB, 5% BM), and *Siderastrea* spp. (15% PB, 5% HB). In March 2017, prevalence increased for *Agaricia* spp. (10% PB), *M. cavernosa* (17% PB) and *Siderastrea* spp. (12% PB, 14% HB), and decreased for *M. hispida* (4% PB, 1.5% HB).

Discussion

The largest WSA reefs suffered a sustained mass coral bleaching episode that spanned across two summers (2016–2017) during the TGBE. Although the positive phase of the 2015–2016 El Niño (Hu and Fedorov 2016, 2017; Rodgers et al. 2017) was followed by a negative phase, high levels of coral bleaching persisted across the Abrolhos reefs into 2017. Our results, besides providing a detailed account of the TGBE effects in the largest WSA reef system, add to the recent evidence that low-diversity turbid-zone reefs may be less susceptible to climate stress

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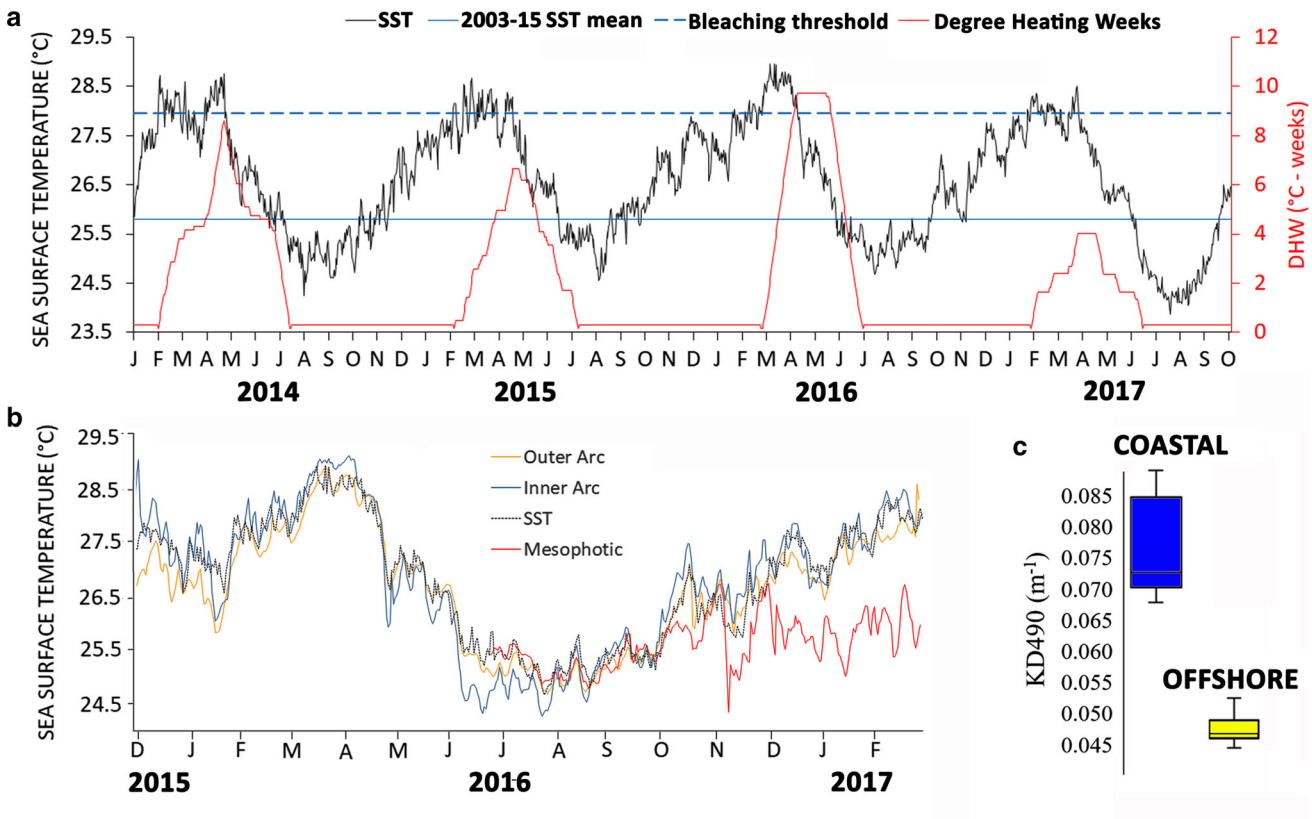

Fig. 3 Recent trends in seawater temperature and turbidity in the Abrolhos reefs, Brazil. **a** Satellite-derived sea surface temperature (SSTs) superimposed in the Degree Heating Weeks (DHW) Index, which indicates accumulated heat stress (2014–2017). Red horizontal line indicates the period when bleached colonies were recorded, solid horizontal blue line indicates decadal average SST, dashed line indicates the bleaching threshold. **b** In situ temperature measurements in the three cross-shelf strata and SSTs (2016–2017) showing lower in situ winter temperatures in the deeper mesophotic reef. **c** Turbidity in coastal and offshore reefs (2014–2017) indexed by the diffuse attenuation coefficient for downwelling irradiance at 490 nm (KD 490) derived from MODIS sensor

than the high-diversity reefs that dwell in clear oligotrophic waters. Peripheral provinces departing from the Caribbean and Indo-Pacific diversity centers, as well as the so-called "marginal" reefs in turbid zones (Morgan et al. 2017), are still unevenly represented in the global bleaching databases (Donner et al. 2017; Hughes et al. 2018a). With a few exceptions (Kelmo and Attrill 2013), the fragmented information from the WSA has low taxonomic and spatial resolution (Castro and Pires 1999). Five bleaching episodes were documented in scattered WSA localities in 1993, 1998, 2003, 2010 and 2016 (see Fig. 2 for reference list), one of them with associated mass coral mortality (Kelmo and Attrill 2013). However, it is likely that there are false negatives (unrecorded events), as coral reef monitoring in Brazil is incipient (Ferreira and Maida 2006), except for the Abrolhos reefs, which have been continuously monitored since 2000 (Amado-Filho et al. 2018).

The 2016–2017 episode in Abrolhos is one of the longest continuous mass bleaching episodes on record. Moderate to severe bleaching occurred from March 2016 to March 2017, and sparse bleached colonies were observed by the resident National Park staff (pers. comm.) in February 2016, and from April to May 2017. Although the 2017 summer thermal anomaly remained below bleaching alert levels, the entire 2014–2017 period was also characterized by low wind speeds, clear skies and calm seas, which may have contributed to the duration of the bleaching episode (see Glynn 1996; Mumby et al. 2001; Skirving et al. 2018). Survival during such an extended mass bleaching may be related to heterotrophic compensation (Anthony 2006; Hughes and Grottoli 2013). Indeed, heterotrophy within the coral assemblage of the turbid-zone Abrolhos reefs, especially among *Mussismilia* spp., seems to be high (Pires et al. 2011; Ribeiro et al. 2018).

Corals with branching morphologies, which often present faster growth and thin soft tissue layers, tend to present high mortalities and are generally the first to bleach (Brown and Suharsono 1990; Glynn 1996; Smith et al. 2014). Their higher surface-to-volume ratios result in increased exposure to warm water and solar radiation, while the thick tissue of most massive forms modulates solar radiation more effectively (Hoegh-Guldberg 1999; Loya et al. 2001).

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Table 2 Percent cover of each coral species in each bleaching category (May 2016, transect data)

Coastal Arc (2 sites)	Top				Wall			
	PB	HB	BM	N	PB	HB	BM	N
AGSP	30.9	14.3	20	42	18.4	34.4	0	139
FAGR	55.2	25.6	1.7	196	20	0	0	8
MADE	NR	NR	NR	0	12.3	0	0	81
MOCA	61.4	18.4	0	146	34.2	60.2	0	389
MUBR	25.4	10.2	0	167	NR	NR	NR	2
MUHA	38.3	25.6	0	122	35.8	13.9	0	63
MUHI	46.9	35.8	0	117	31.1	13.6	0	86
POSP	47.6	33.8	0	57	50.8	13.3	0	24
SISP	45.2	28.2	0.2	298	57.1	10.8	5.2	118
MISP	15.2	43.2	14.9	43	NR	NR	NR	0
Offshore Arc (2 sites)	Top				Wall			
	PB	HB	BM	N	PB	HB	BM	N
AGSP	20.8	1.5	0	160	11.8	74.5	9	96
FAGR	71.5	14	1.9	110	50	0	0	2
MADE	NR	NR	NR	0	31	0	0	29
MOCA	69.4	14.2	0	132	39	37.3	0	84
MUBR	19.4	3.1	0	156	NR	NR	NR	0
MUHA	41.8	50.5	0	70	NR	NR	NR	1
MUHI	60.6	31.8	0	91	33	19.7	0	92
POSP	10.1	87.3	0	94	5.2	38.5	0	32
SISP	2.7	85.1	0.5	444	33.2	0	0	62
MISP	48.8	31.9	11.6	90	NR	NR	NR	0

Prevalence indicated with colors. Red: > 40.1%, Orange: 30.1–40%, Yellow: 20.1–30%, Blue: 10.1–20%, Green: 0.1–10%, Gray: 0%, N: number of colonies, NR: not recorded. *AGSP*, *Agaricia* spp.; *FAGR*, *Favia gravida*; *MADE*, *Madracis decactis*; *MOCA*, *Montastraea cavernosa*; *MUBR*, *Mussismilia braziliensis*; *MUHA*, *M. harttii*; *MUHI*, *M. hispida*; *POSP*, *Porites* spp.; *SISP*, *Siderastrea* spp.; *MISP*, *Millepora* spp.; *PB*, Partially Bleached; *HB*, Heavily Bleached; *BM*, Bleaching and Mortality (see "Materials and Methods")

Branching hydrocorals *Millepora* spp. were among the most affected species in Abrolhos, and also in northern Bahia during the 1997–1998 event (Kelmo and Attrill 2013). Indeed, milleporans may even reach regional extirpation after temperature extremes (e.g., Glynn and de Weerdt 1991; Smith et al. 2014). In the WSA, decrease of milleporans (4 species) can have severe ecosystem-level consequences, as they are the region's only branching corals and provide important nursery habitat for several reef fishes (Coni et al. 2013). Besides milleporans, other severely affected species on pinnacles tops (> 40% of colonies bleached) include massive *Porites* spp., *F. gravida*, and *Siderastrea* spp., all of which present very thin tissues (authors' observations). Plate-like forms (e.g., *Agaricia* spp.), typical of low-light environments such as pinnacle's walls, also tend to be fast-growing, have relatively thin tissues, and to be more susceptible to bleaching (Loya et al. 2001). Coral morphologies can be highly plastic (Forsman et al. 2009), and there are clear zonation patterns of species and/or growth forms related to collinear variations of depth, turbulence and light gradients (Graus and Macintyre 1989).

The Brazilian-endemic and endangered coral *M. braziliensis* comprises one of the most abundant species on pinnacles' tops (Francini-Filho et al. 2008, 2013), and was one of the less susceptible species. This massive species with a thick soft tissue layer occurs in an extremely narrow geographic (13–20°S) and depth range (1–15 m) and seems to be highly adapted to the local conditions of the Abrolhos' pinnacles' tops (Pires et al. 2011). However, most other Abrolhos' coral species presented significant between-habitat differences in abundance and bleaching prevalence. Although shading may provide protection against thermal stress (e.g., Coelho et al. 2017), we found that species typical of pinnacles' walls, such as *M. cavernosa* and *Agaricia* spp. (the 1st and 2nd most abundant corals in this habitat), presented lower bleaching prevalence in the tops, where they are less abundant. Therefore, under extreme scenarios, the few survivors in habitats where abundance and growth rates are low may be a critical component in reef resilience, especially when mortality is high (Hoogenboom et al. 2017; Hughes et al. 2018b).

Depth and cross-shelf trends of bleaching prevalence and severity may be variable (Marshall and Baird 2000;

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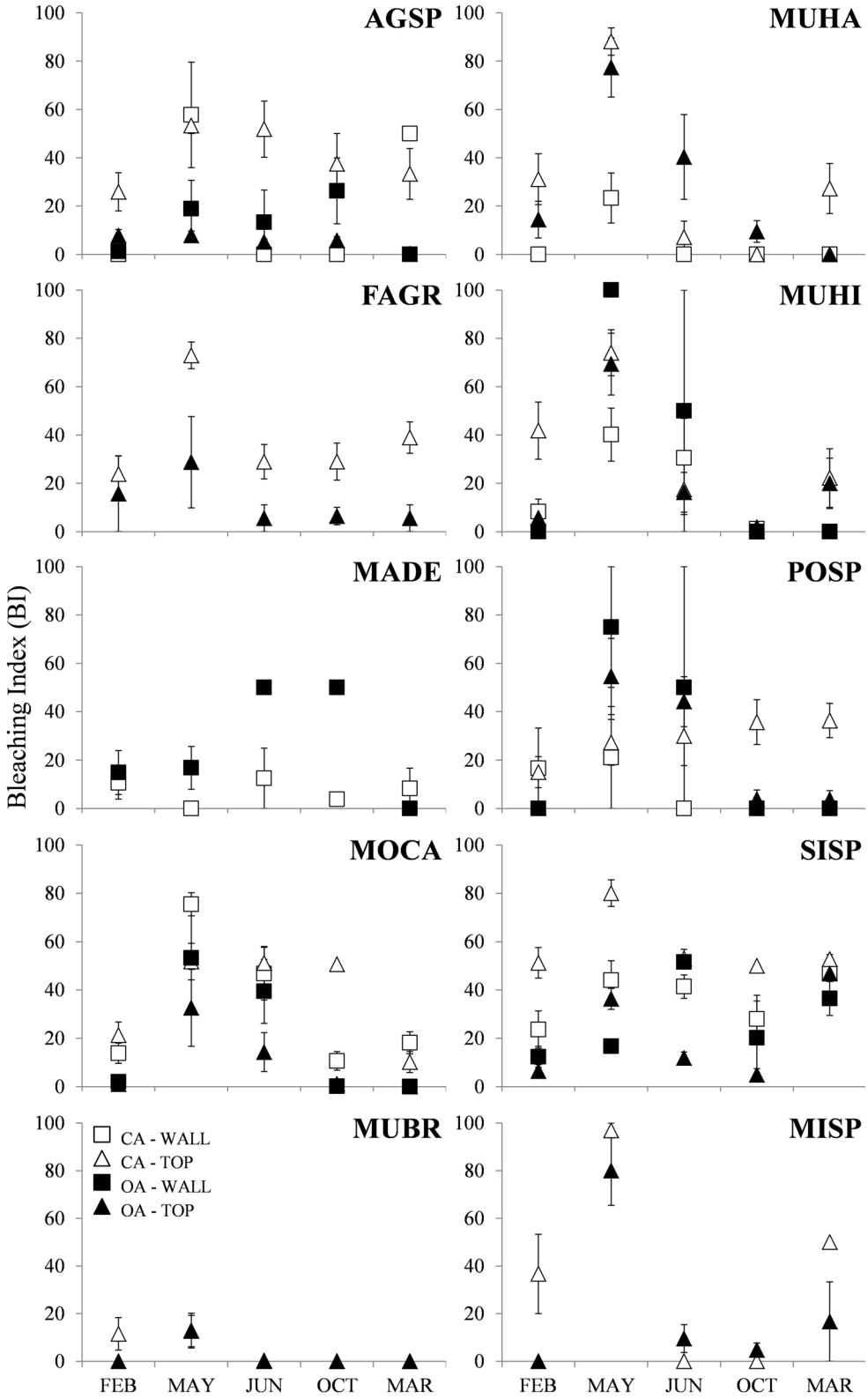

Fig. 4 Temporal variation (February 2016–March 2017) in the Bleaching Index (BI) for the 10 most abundant coral species in the different habitats and cross-shelf strata of the Abrolhos Reefs. *AGSP*, *Agaricia* spp.; *FAGR*, *Favia gravida*; *MADE*, *Madracis decactis*; *MOCA*, *Montastraea cavernosa*; *MUBR*, *Mussismilia braziliensis*; *MUHA*, *M. harttii*; *MUHI*, *M. hispida*; *POSP* *Porites* spp.; *SISP*, *Siderastrea* spp.; *MISP*, *Millepora* spp. Bars represent Standard Errors

Berkelmans et al. 2004), and turbid-zone reefs may present high bleaching tolerance (Le Nohaïk et al. 2017; Morgan et al. 2017). In Abrolhos, with the exception of low bleaching levels in the deeper mesophotic reefs, we did not find a cross-shelf trend in bleaching prevalence, and interactions between arc and habitat effects were overall

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Table 3 Results of the PERMANOVA tests with the Bleaching Index data, contrasting sites and habitats

Source	Arc					Habitat				
	df	SS	MS	F	*p*	*df*	SS	MS	F	*p*
AGSP	1	394.1	394.1	0.459	ns	1	7547.5	7547.5	87.804	**
FAGR	1	58.242	58.2	0.223	ns	1	8357.3	8357.3	32.048	***
MADE	1	381.15	381.2	0.700	ns	1	2077.8	2077.8	38.181	*
MOCA	1	478.09	478.1	11.056	ns	1	1452.5	1452.5	33.589	ns
MUBR	1	447.2	447.2	26.757	ns	1	731.44	731.44	43.764	*
MUHA	1	29.722	29.7	720	ns	1	7910.4	7910.4	19.152	***
MUHI	1	106.85	106.9	0.270	ns	1	3496	3496	88.328	**
POSP	1	371.52	371.5	0.331	ns	1	2923	2923	2.605	0.1053
SISP	1	892.91	892.9	1.523	ns	1	1792.7	1792.7	30.578	0.0944
MISP	1	215.82	215.8	0.691	ns	1	14,835	14,835	47.463	***

Source	Arc × Habitat					Residual			Total	
	df	SS	MS	F	*p*	*df*	SS	MS	*df*	SS
AGSP	1	3320.4	3320.4	38.628	ns	14	12034	859.6	17	22306
FAGR	1	751	751	0.000288	ns	14	3650.8	260.8	17	12165
MADE	1	381.2	381.2	0.7	ns	14	7618.8	544.2	17	10289
MOCA	1	456.6	456.6	10.559	ns	14	6054	432.4	17	8648.3
MUBR	1	43	43	2.56E−01	ns	14	2339.9	167.1	17	3528
MUHA	1	3800	3800	92.002	**	14	5782.5	413.0	17	16435
MUHI	1	19.7	19.7	499	ns	14	5541.2	395.8	17	9266.8
POSP	1	2889.3	2889.3	2.575	ns	14	15709	1122.1	17	21312
SISP	1	329.6	329.6	0.562	ns	14	8208	586.3	17	11077
MISP	1	215.82	215.82	0.690	ns	14	4375.8	312.6	17	19428

AGSP, *Agaricia* spp.; *FAGR*, *Favia gravida*; *MADE*, *Madracis decactis*; *MOCA*, *Montastraea cavernosa*; *MUBR*, *Mussismilia braziliensis*; *MUHA*, *M. harttii*; *MUHI*, *M. hispida*; *POSP*, *Porites* spp.; *SISP*, *Siderastrea* spp.; *MISP*, *Millepora* spp. *df*, degrees of freedom; *SS*, sum of squares; *MS*, mean sum of squares; *F*, pseudo-F; *ns*, non-significant

$^{*}p < 0.05$; $^{**}p < 0.01$; $^{***}p < 0.001$

weak and restricted to *Agaricia* spp. and *M. harttii* (see Table 3). We did not record significant temperature differences between coastal and offshore reefs (Fig. 3), and the strong habitat effect due to the oddly shaped pinnacles' morphology seems to overwhelm the weaker cross-shelf effect.

Quadrat data was too limited to allow statistically robust spatial comparisons, but bleaching prevalence in the deeper mesophotic California Reef tended to be lower than 10%, i.e., less than in the emergent reefs (> 30%). Offshore, water column stratification seemed to intensify in the summer, and our in situ temperatures from the California Reef measured at 20 m depth indeed deviate sharply from satellite-SSTs. Mesophotic reefs can be a thermal refuge, with low bleaching and bleaching related mortality, but less than half of the species recorded in the pinnacles' reefs also occur in the deeper mesophotic California Reef, which also contained more limited coral cover. In addition, bleaching threshold temperatures at California Reef are likely lower than in shallow reefs (Smith et al. 2016). Therefore, the Abrolhos' deeper reefs only provide a partial refuge to climate changes (Bongaerts et al. 2010; Smith et al. 2016).

Bleaching levels in Abrolhos were high (up to 73% of colonies bleached in offshore arc pinnacles tops), but overall mortality was low (< 3% of total coral cover) as of March 2017, contrasting with the high mortality rates recorded in the Indo-Pacific and the Caribbean, with > 80% coral mortality even in undisturbed central and western Pacific Reefs (Rodgers et al. 2017; Hughes et al. 2017).

Akin to coral reefs in regions with extreme temperatures (Krueger et al. 2017), turbid-zone reefs dominated by stress-tolerant species may function as climate-change refugia (van Woesik et al. 2012; Morgan et al. 2017), either due to reduced irradiance, species composition, or local

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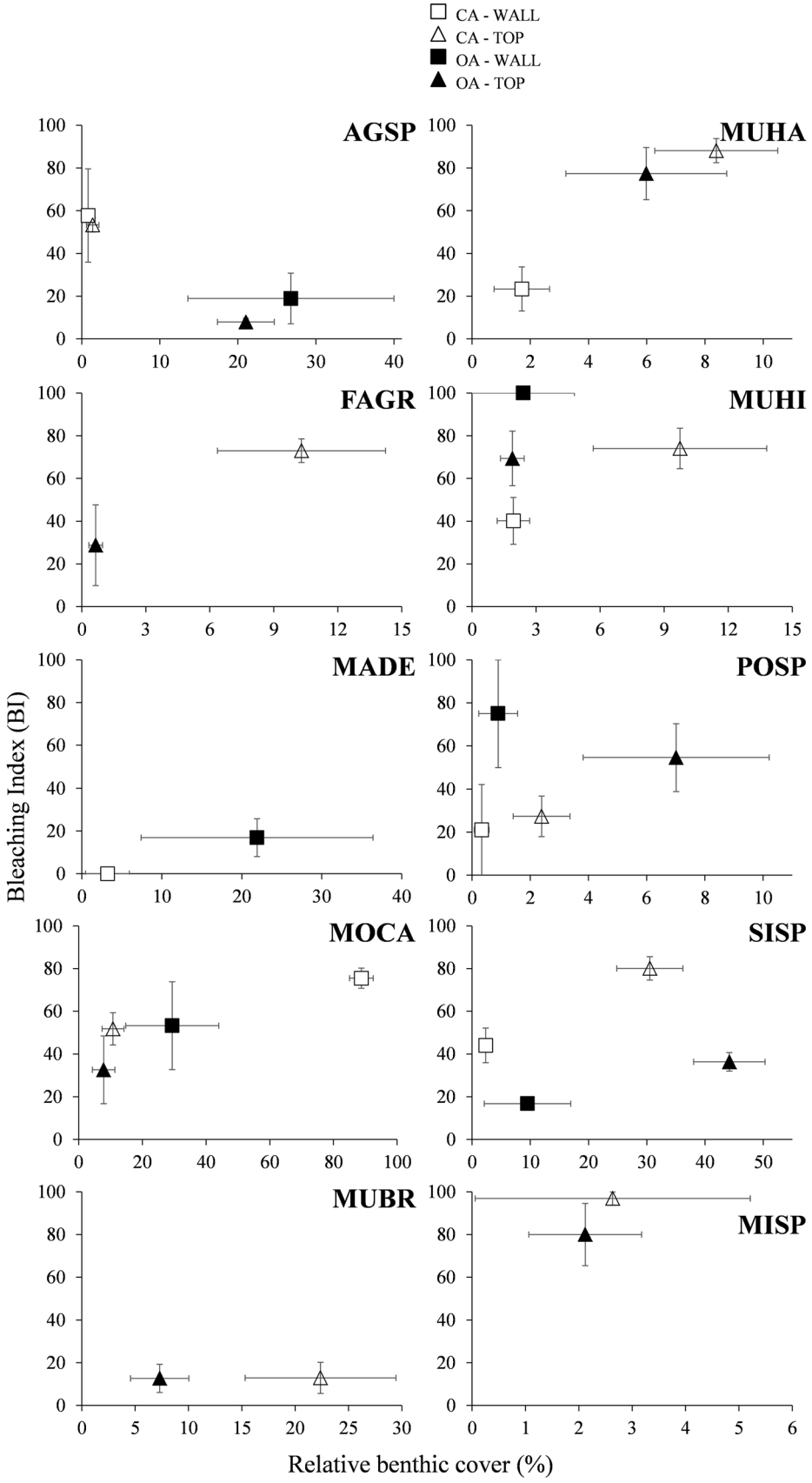

Fig. 5 Bleaching Index (BI) and relative cover of the most abundant coral species in both habitats (pinnacles tops and walls) of the coastal and offshore arcs during the event's peak in May 2016. *AGSP*, *Agaricia* spp.; *FAGR*, *Favia gravida*; *MADE*, *Madracis decactis*; *MOCA*, *Montastraea cavernosa*; *MUBR*, *Mussismilia braziliensis*; *MUHA*, *M. harttii*; *MUHI*, *M. hispida*; *POSP*, *Porites* spp.; *SISP*, *Siderastrea* spp.; *MISP*, *Millepora* spp. *CA* coastal arc; *OA* offshore arc. Bars represent Standard Errors

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adaptation/acclimation. However, these "nearshore climate-change refugia" are not immune to bleaching (Le Nohaïk et al. 2017; Morgan et al. 2017), especially under the disturbing scenario forecasted for the twenty-first century (Hughes et al. 2018a). Indeed, the loss of a few coral species that are more susceptible to thermal anomalies (e.g., *Millepora* spp.) may compromise such low diversity systems with low functional redundancy.

Acknowledgements This work is a contribution from the Abrolhos Network (www.abrolhos.org) and was co-funded by Brazil's National Research Council (CNPq) through the Long Term Ecological Monitoring Program (PELD–Site ABRS), the International Ocean Discovery Program (CAPES), and ANP/Brasoil. ACB, FVR, FSM, GHPF, GMAF, LMN, MK, PSS and RLM acknowledge individual grants from CNPq, CAPES, FAPESP and FAPERJ. We thank the Abrolhos National Marine Park staff, especially Fernando Repinaldo and Berna Barbosa, for research permits, field support and information sharing. Two anonymous reviewers provided helpful input.

Compliance with ethical standards

Conflict of interest The authors declare no conflict of interest.

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